# REZAUR RASHID, PhD

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## Education

- **The University of North Carolina at Charlotte, Charlotte, NC, USA** *2019 – 2024*
  Ph.D. in Computer Science | GPA: 3.90/4.00
  Dissertation Title: *Beyond Causal Pairs: A Probabilistic Approach to Causal Structure Learning from Cause-Effect Pair Relationships Using Graph Neural Network*.
  Advisor: Gabriel Terejanu, PhD.
- **BRAC University, Dhaka, Bangladesh** *2013 –2017*
  B.Sc. in Computer Science and Engineering | GPA: 3.67/4.00 (High Distinction Honors)

## Research Interests

Explainable AI, causal AI, data science, large language models, and graph neural networks

## Research Projects

- **Causal Discovery and Representation Learning** [9], [15-17], [25], [29]
  Developed probabilistic graph-based methods for causal structure learning and causal feature selection, with applications to biomedical and social data.
- **Explainable AI for Cancer Treatment and Outcomes** [3], [7], [13], [23-24], [29]
  Built interpretable machine learning, graph neural network, and causal inference models for radiation treatment interruption and cancer outcomes using clinical and social determinants data.
- **Multimodal Clinical AI Systems and Data Integration** [2], [6-7], [12], [21]
  Designed X-CART, an AI-ready architecture integrating structured EHR data and clinical narratives; and extended this framework toward a radiation oncology digital-twin model.
- **Empathic and Agentic AI for Patient Simulation** [4-5], [8], [10], [22], [27], [28]
  Developed synthetic patient personas and dialogue evaluation pipelines to assess empathy, safety, and clinical adequacy in AI-supported oncology navigation.
- **Large-Scale Drug–Condition Association Discovery**
  Applied distributed data processing, temporal exposure windows, and association measures including PMI, NPMI, and lift to identify candidate off-label medication–condition relationships (around 146 million).
- **Computational Social Science and Health Sentiment Dynamics** [1], [11], [14]
  Studied polarization, peer influence, and health-related sentiment using causal inference, graph methods, and large language models.
- **Causal Analytics: Modeling the "Why" in Business Operations, Lowe's Companies, Inc.**
  Developed causal reasoning and automated feature-generation methods for explainable supply-chain and operational analytics.

## Awards and Honors

- Graduate Travel Funding for academic and professional conferences, GPSG, UNC Charlotte 2022, *2024*
- Full Ph.D. Scholarship from Graduate Assistant Support Plan (GASP), UNC Charlotte *2019 – 2024*
- Graduate School Summer Fellowship (GSSF) Award, UNC Charlotte *2023*
- Vice-chancellor's list of honors for academic excellence, BRAC University *2013, 2016*
- Dean's list of honors for academic excellence, BRAC University *2015, 2016*

## Publications

- **Journal Papers**

[1] Rashid R, White BM, Kumsa FA, and Shaban-Nejad A. "Benchmarking Large Language Models for Sentiment Analysis in Pregnancy-Related Weight Discussions on Reddit: A Human Performance-Aligned Evaluation". *JMIR Form. Res.* (Accepted and in press). 2026.

[2] Zink JA, Rashid R, Chinthala L, Kheirinejad S, Rahimabad PK, Hashtarkhani S, Kumsa FA, Brett CL, Davis RL, Schwartz DL, and Shaban-Nejad A. "X-CART: A Scalable Multimodal Data Architectural Blueprint for Explainable AI-Driven Patient Navigation in Radiation Oncology". *ESMO Real World Data Digit Oncol, 13.* 2026; p.100771.

[3] Hashtarkhani S, Brett CL, Kumsa FA, Rashid R, Rahimabad PK, Kheirinejad S, Davis RL, Schwartz DL, and Shaban-Nejad A. "Psychosocial Distress and Risk of Radiation Treatment Interruptions: Integrating Patient-Reported Outcomes, Electronic Health Records, and Machine Learning". *JMIR Cancer.* 2026 Oct 19; 12:e92899.

[4] Kheirinejad S, White BM, Rahimabad PK, Zink JA, Hashtarkhani S, Kumsa FA, Rashid R, Chinthala L, Brett CL, Davis RL, Schwartz DL, and Shaban-Nejad A. "User-Centered Design for Digital Patient-Navigation Tools in Oncology: A Scoping Review". *JMIR Hum Factors.* 2026 Apr 29; 13:e87686.

[5] White BM, Aitken GL, Zink JA, Rahimabad PK, Kumsa FA, Hashtarkhani S, Rashid R, Kheirinejad S, Girdwood T, Brett CL, Davis RL, Schwartz DL, and Shaban-Nejad A. "Empathic AI for Patient-Centered Cancer Care: A Scoping Review of Patient Navigation, Support, and Clinical Practice". *JMIR Cancer.* 2026 Apr 9; 12:e82336

[6] Hashtarkhani S, Rashid R, Brett CL, Chinthala L, Kumsa FA, Zink JA, Davis RL, Schwartz DL, and Shaban-Nejad A. "Cancer diagnosis categorization in electronic health records using large language models and BioBERT: Model performance evaluation study". *JMIR cancer*. 2025 Oct 2; 11:e72005.

[7] Kumsa FA, Brett CL, Hashtarkhani S, Rashid R, Chinthala L, Zink JA, Davis RL, Shaban-Nejad A, and Schwartz DL. "Leveraging Centralized Health System Data Management and Large Language Model–Based Data Preprocessing to Identify Predictors for Radiation Therapy Interruption". *JCO Clin Cancer Inform, 9*. 2025; p.e2500218.

- **Peer-reviewed Conference Papers**

[8] Ammar R, Rashid R, Kheirinejad S, Mote T, Schwartz DL, and Shaban-Nejad A. "Clinical Communication Complexity and AI Navigation Performance in Simulated Oncology Conversations: A Pilot Study". *EFMI STC 2026, Athens, Greece* (Accepted). 2026

[9] Rashid R, and Terejanu G. "From Observations to Causations: A GNN-Based Probabilistic Prediction Framework for Causal Discovery". In *Proceedings of the 17th International Joint Conference on Knowledge Discovery, Knowledge Engineering and Knowledge Management - Volume 1: KDIR.* 2025 Oct; (pp. 337-348). SciTePress.

[10] Rashid R, Kheirinejad S, White BM, Hashtarkhani S, Rahimabad PK, Kumsa FA, Chinthala L, Zink JA, Brett CL, Davis RL, Schwartz DL, and Shaban-Nejad A. "Simulating Empathic Interactions with Synthetic LLM-Generated Cancer Patient Personas". In *Good Evaluation-Better Digital Health*. 2025 Oct; (pp. 72-76). IOS Press.

[11] Chowdhury J, Rashid R, and Terejanu G. "Measuring Social Media Polarization Using Large Language Models and Heuristic Rules". In: *An, A, Cuzzocrea, A, Hu, H. (eds) Social Networks Analysis and Mining. FAB 2025.* Lecture Notes in Computer Science. 2025 Aug; vol 16324. Springer, Cham.

[12] Rashid R, Hashtarkhani S, Kumsa FA, Chinthala L, White BM, Zink JA, Brett CL, Davis RL, Schwartz DL, and Shaban-Nejad A. "AI-Ready Multimodal Data Pipeline to Enrich Cancer Care". In *2024 IEEE International Conference on Big Data (BigData).* 2024 Dec; (pp. 8814-8818). IEEE.

[13] Hashtarkhani S, White-Means S, Li S, Rashid R, Kumsa, F, Lemon C, Chipman, L, Dapremont J, White B, and Shaban-Nejad A. "An Explainable AI Data Pipeline for Multi-Level Survival Prediction of Breast Cancer Patients Using Electronic Medical Records and Social Determinants of Health Data". In *2024 IEEE International Conference on Big Data (BigData).* 2024 Dec; (pp. 8661-8664). IEEE.

[14] Rashid R, Melton J, Ghorbani O, Krishnan S, Reid S, and Terejanu G. "Quantifying Influencer Impact on Affective Polarization". In *2024 International Conference on Machine Learning and Applications (ICMLA).* 2024 Dec; (pp. 1135-1140). IEEE.

[15] Rashid R, Chowdhury J, and Terejanu G. "Causal feature selection: Methods and a novel causal metric evaluation framework". In *2023 IEEE 10th International Conference on Data Science and Advanced Analytics (DSAA)*. 2023 Oct; (pp. 1-9). IEEE.

[16] Chowdhury J, Rashid R, and Terejanu G. "Evaluation of Induced Expert Knowledge in Causal Structure Learning by NOTEARS". In *Proceedings of the 12th International Conference on Pattern Recognition Applications and Methods - Volume 1: ICPRAM*. 2023 Fab; (pp. 136-146). SciTePress.

[17] Rashid R, Chowdhury J, and Terejanu G. "From causal pairs to causal graphs". In *2022 21st IEEE International Conference on Machine Learning and Applications (ICMLA)*. 2022 Dec; (pp. 802-807). IEEE.

[18] Rashid MR, Shuvra NKD, Roy RS, and Sarker S. "A Theoretical Approach of Fall Control System for Disabled People Using Double Inverted Pendulum". In *2018 4th International Conference on Electrical Engineering and Information & Communication Technology (iCEEiCT)*. 2018 Sep; (pp. 49-53). IEEE.

[19] Rashid MR, Uzzaman N, Hossain S, and Shuvra NKD. "Development of a self-navigating algorithm for library book finder robot". In *2017 3rd International conference on electrical information and communication technology (EICT)*. 2017 Dec; (pp. 1-6). IEEE.

[20] Uzzaman N, Hossain S, Rashid R, and Hossain A. "A comparative analysis between light dependent and ultrasonic method of gesture recognition". In *2016 3rd International Conference on Electrical Engineering and Information Communication Technology (ICEEICT)*. 2016 Sep; (pp. 1-5). IEEE.

- **Peer-reviewed Abstracts and Workshop Papers**

[21] Rashid R, Kheirinejad S, Rahimabad PK, Hashtarkhani S, White BM, Kumsa FA, Chinthala L, Brett CL, Davis RL, Schwartz DL, and Shaban-Nejad A. "Simulating Radiation Therapy Interruptions Using AI-Generated Digital Twins: A Literature-Calibrated Synthetic Cohort". *ESMO Real World Data Digit Oncol. 2026;* (Accepted). Abstract 291.

[22] Rashid R, Kheirinejad S, Rahimabad PK, White BM, Hashtarkhani S, Kumsa FA, Chinthala L, Zink JA, Brett CL, Davis RL, Schwartz DL, and Shaban-Nejad A. " Preclinical In Silico-Human Evaluation of AI-Based Cancer Patient Navigation in Radiotherapy". *ESMO Real World Data Digit Oncol. 2026;* (Accepted). Abstract 292.

[23] Rashid R, Hashtarkhani S, Rahimabad PK, White BM, Kumsa FA, Chinthala L, Zink JA, Brett CL, Davis RL, Schwartz DL, and Shaban-Nejad A. "Abstract A061: Machine Learning and Causal Inference-Based Predictive Risk Modeling of Unplanned Radiation Treatment Interruption". *Clinical Cancer Research*, *31*(13_Supplement). 2025; pp.A061-A061.

[24] Hashtarkhani S, Rashid R, Rahimabad PK, Kumsa FA, White BM, Chinthala L, Zink JA, Brett CL, Davis RL, Schwartz DL, and Shaban-Nejad A. "Abstract A062: Uncovering Social and Clinical Determinants of Baseline Distress Prior to Radiation Therapy: An Explainable Machine Learning Approach". *Clinical Cancer Research*, *31*(13_Supplement). 2025; pp.A062-A062.

[25] Rashid R, and Terejanu G. "Graph neural networks for probabilistic causal discovery". In *Proceedings of the 9th Causal Inference Workshop at UAI 2024*. 2024.

- **Manuscripts Under Review and In Revision**

[26] Rashid R, Hashtarkhani S, White BM, Kumsa FA, and Shaban-Nejad A. "Agentic AI in Healthcare: A Narrative Review of Architectures and Emerging Applications". *JMIR AI.* (Submitted). 2026

[27] Rashid R, Kheirinejad S, Rahimabad PK, Hashtarkhani S, White BM, Kumsa FA, Zink JA, Chinthala L, Brett CL, Davis RL, Schwartz DL, and Shaban-Nejad A. "Stress-Testing Empathic AI Navigation for Radiation Therapy Interruptions Using Synthetic Patient Dialogues". *ESMO Real World Data Digit Oncol.* (Submitted). 2026.

[28] White BM, Rahimabad PK, Kheirinejad S, Zink JA, Rashid R, Kumsa FA, Hashtarkhani S, Girdwood T, Brett CL, Davis RL, Schwartz DL, and Shaban-Nejad A. "Agentic AI in Oncology: A Scoping Review and Conceptual Framework for Clinical Integration". *NPJ Precis. Oncol.* (Revise and resubmit). 2026.

[29] Rashid R, Rahimabad PK, Hashtarkhani S, Kheirinejad S, White BM, Kumsa FA, Chinthala L, Zink JA, Brett CL, Davis RL, Schwartz DL, and Shaban-Nejad A. "Explainable AI With Graph Neural Networks and Causal Inference for Predicting Unplanned Radiation Therapy Interruptions". *IEEE Journal of Biomedical and Health Informatics* (Revise and resubmit). 2025.

## Work Experience

- **Postdoctoral Research Fellow,** Center of Biomedical Informatics (CBMI) *September 2024 – Present*
  Department of Pediatrics, College of Medicine
  The University of Tennessee Health Science Center, Memphis, TN, USA
- **Graduate Research Assistant,** Department of Computer Science *August 2019 – August 2024*
  The University of North Carolina at Charlotte, Charlotte, NC, USA
- **Graduate Intern**, Data Science *July 2022 – August 2022*
  Lowe's Charlotte Technology Hub, Lowe's Companies, Inc, Charlotte, NC, USA

## Teaching Experience

- **Guest Lecturer**, IPEH 811: AI in Med. and Health Research *Fall 2026*
  The University of Tennessee Health Science Center, Memphis, TN, USA
  - Delivered a two-hour invited lecture, "*AI in Population Health: Mapping Geography, Access, and Care Delivery*" on geospatial health analytics, access to care, clinical AI, multimodal data, large language models, and responsible evaluation.
- **Graduate Teaching Assistant**, Department of Software Information Systems *Summer 2020*
  The University of North Carolina at Charlotte, Charlotte, NC, USA
  - Co-taught Network-Based Application Development (IT IS 4166/5166) course to 32 students.
  - Adapted teaching methods (Remote working) to virtual environments due to COVID-19 pandemic ensuring uninterrupted and quality education.
- **Lecturer,** Department of Computer Science and Engineering *May 2017 – June 2019*
  Bangladesh University, Dhaka, Bangladesh
  - **Courses Taught**: Algorithms and Data Structures, Pattern Recognition, Artificial Intelligence.
  - Fostered critical thinking and problem-solving skills among students.

## Research Mentoring and Supervision

- **Krutik Nigam**, Medical Student Researcher, UTHSC *Summer 2026 – Present*
  - Mentoring research extending explainable AI models for radiation therapy interruptions toward intervention-oriented decision support, including intervention mapping, perturbation simulations, and explanation-guided analysis of potentially modifiable risk factors.
- **Raneem Ammar**, Medical Student Research Fellow, UTHSC *Summer 2026*
  - Mentored research on communication complexity and AI performance in synthetic oncology navigation dialogues, resulting in a student-led paper accepted at *EFMI STC 2026*, Athens, Greece.
- **Thrisha Mote**, Medical Student Research Fellow, UTHSC *Summer 2026*
  - Mentored research comparing automated dialogue metrics with expert clinical ratings, resulting in a student-led manuscript prepared for submission to *JMIR Cancer Research Letter*.
- **Three High School Students**, ENRICH Summer Program, CBMI, UTHSC *Summer 2026*
  - Mentored three high school students in designing and building a modular, AI-powered research platform for the lab, using locally run large language models.
- **Miguel**, Graduate Student, Technical University of Denmark (DTU) *Spring 2025*
  - Co-mentored research using variational autoencoders to identify latent metabotypes in UK Biobank metabolomics data.
- **Sai K. Nandipati**, Graduate Student, UTHSC *Fall 2024*
  Co-mentored research on synthetic data generation and causal structure learning for AI model development and evaluation.

## Selected Invited Talks and Presentations

- ***From Pilot to Practice:* Explainable AI for Cancer Care** September 2026
  Tennessee Statewide Cancer Collaborative, Nashville, TN
  - Presented the UTHSC-Dell-AHEAD collaboration for operationalizing explainable AI in radiation oncology, including RTI risk prediction, clinical decision support, and a roadmap for hospital pilot deployment.

## Professional Service

- **Journal Reviewer:** *Front. Commun.* (2026), *IEEE Trans. Neural Netw. Learn. Syst.* (2026), *Comp. Struct. Biotechnol. J.* (2026), *Neurocomputing* (2025)
- **PC Member:** ASONAM (2024-2025)
- **Conference Reviewer:** WCCI/IJCNN (2023-2026)

## Professional Skills

- **Machine Learning and AI:** Deep learning; supervised and unsupervised learning; causal modeling; graph neural networks; large language models.
- **Statistical Analysis:** Regression modeling; hypothesis testing; A/B testing; ANOVA; causal inference methods including difference-in-differences, propensity score matching, and interrupted time series analysis.
- **Programming and Data Science:** Python, R, SQL/MySQL, Java, MATLAB; data analysis and visualization; scientific computing.
- **Framework and Libraries:** PyTorch, TensorFlow/Keras, Scikit-learn, SciPy, Pandas, Dask, Modin, NumPy, Matplotlib, Seaborn, NetworkX, XGBoost, PyTorch Geometric (PyG), Deep Graph Library (DGL), PyCaret, PySpark, NLTK.
- **Systems and Infrastructure:** High-performance computing (HPC); distributed data processing; containerization with Docker; cloud-based workflows.
- **Tools and Platforms:** Git, Bitbucket; JupyterLab, RStudio; LaTeX, Overleaf, R Markdown, MS Office.
- **Operating Systems:** Windows, macOS, Linux (working knowledge).

## Open-Source Python Library

- Causal Feature Selection [GitHub]
- Automated Feature Synthesis [GitHub]
- RTI Empathic AI Simulation [GitHub]

## Other Activities

- **Treasurer** for Muslim Student Association (MSA), UNC Charlotte, Charlotte, NC *2020 – 2021*
- **Senator** for Bangladeshi Student Organization, UNC Charlotte, Charlotte, NC 2019 – *2020*